

Islamic University

Faculty of Engineering

Department of Computer Eng. Tech.

Year No. Forth



الجامعة الإسلامية

كلية الهندسة

قسم هندسة تقنيات الحاسوب

المرحلة الرابعة

Name of lab.: Information Theory and Coding

اسم المختبر: نظرية المعلومات والترميز

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Lab. 1

2026-2027

Description of Experiment / Practice

1. Experiment / Practice No: 1

2. Title of Experiment / Practice: Self-Information Measurements

3. Objectives of Experiment / Practice:

- Understand the concept of self-information.
- Study the relationship between probability and information.
- Calculate the self-information of different events.
- Use MATLAB to calculate self-information values.
- Analyze how event probability affects the amount of information.

4. Number of Required Hours: (2 hours)

5. Expected Outcomes of Experiment / Practice

After completing this experiment, students will be able to:

- Define self-information.
- Calculate the information content of an event.
- Explain the relationship between probability and information.
- Compare the information content of different events.
- Use MATLAB to calculate self-information.

6. Language: English

7. Short Explanation (theory)

What is Self-Information?

In Information Theory, self-information measures the amount of information obtained when an event occurs. The amount of information depends on the probability of the event. The self-information of an event x is defined as:

$$I(x) = -\log_2 P(x)$$

$$0 < P(x) \leq 1, \text{ For self - information}$$

Where:

- $I(x)$ = Self-information of event x
- $P(x)$ = Probability of event x

- The unit of information is bit

Basic Idea

The less probable an event is, the more information it provides.

$$P(x) \downarrow \rightarrow I(x) \uparrow$$

Similarly:

$$P(x) \uparrow \rightarrow I(x) \downarrow$$

So:

Rare event $\rightarrow$ More information

Common event $\rightarrow$ Less information

Example 1

If an event has probability:

$$P(x) = \frac{1}{2}$$

Then:

$$I(x) = -\log_2 \left(\frac{1}{2} \right)$$

$$I(x) = 1 \text{ bit}$$

Example 2

If an event has probability:

$$P(x) = \frac{1}{8}$$

Then:

$$I(x) = -\log_2 \left(\frac{1}{8} \right)$$

$$I(x) = 3 \text{ bits}$$

Therefore, the event with probability $1/8$ carries more information than the event with probability $1/2$.

Comparison of Different Probabilities

Probability $P(x)$	Self-Information $I(x)$
1	0 bit
0.5	1 bit

0.25	2 bits
0.125	3 bits

8. Required Materials

- MATLAB Software
- Computer/Laptop
- MATLAB Symbolic Math Toolbox
- Lecture Notes

9. Apparatus and Instruments

- Personal Computer
- MATLAB Installed Environment

10. Procedures

1. Start MATLAB.
2. Clear the command window, workspace variables, and previously opened figures.
3. Define a set of event probabilities.
4. Calculate the self-information for each probability using:

$$I(x) = -\log_2 P(x)$$

5. Display the probability and corresponding self-information values.
6. Compare the information values of high-probability and low-probability events.
7. Plot probability against self-information.
8. Observe how self-information changes as probability decreases.

MATLAB Code:

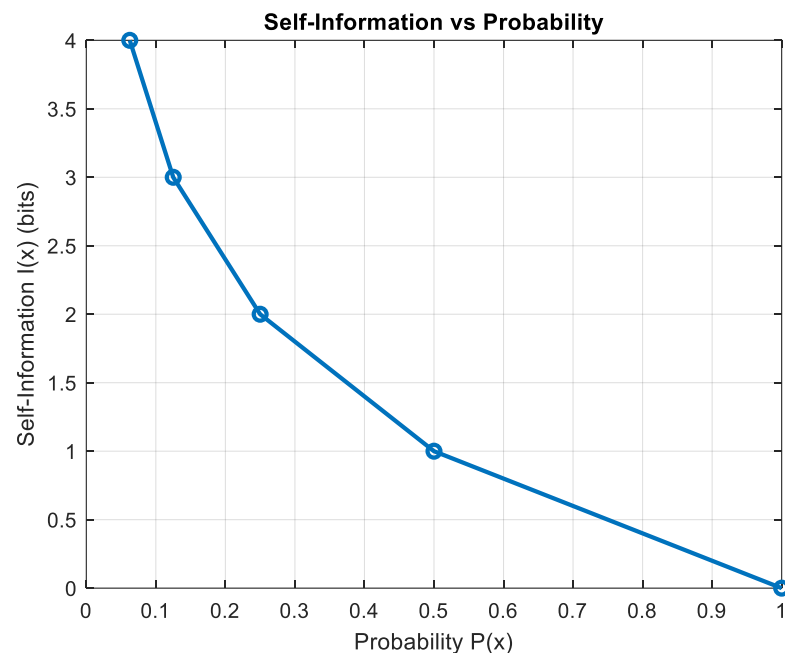
```
clc; clear; close all;  
P = [1 0.5 0.25 0.125 0.0625];  
I = -log2(P);  
disp('Probability    Self-Information (bits)')  
disp([P' I'])  
figure  
plot(P,I,'o-','LineWidth',2)  
grid on  
xlabel('Probability P(x)')  
ylabel('Self-Information I(x) (bits)')  
title('Self-Information vs Probability')
```

9. Observation / Result

The calculated values are:

Command Window	
Probability	Self-Information (bits)
1.0000	0
0.5000	1.0000
0.2500	2.0000
0.1250	3.0000
0.0625	4.0000

fx >> |



The results show that self-information increases when the probability of an event decreases.

Therefore:

Lower Probability $\Rightarrow$ Higher Information

10. Conclusion

In this experiment, self-information was calculated for different event probabilities using MATLAB. The results demonstrate that the amount of information associated with an event depends on its probability. Events with lower probabilities provide more information, while events with higher probabilities provide less information. The self-information is calculated using:

$$I(x) = -\log_2 P(x)$$

and its unit is bit.

11. Homework

Q1: Using MATLAB, calculate and plot the self-information for the following probabilities:

$P(x) = 0.5, 0.25, 0.1, 0.01$.

12. References

- Claude E. Shannon, A Mathematical Theory of Communication.
- Thomas M. Cover and Joy A. Thomas, Elements of Information Theory.
- MATLAB Documentation.